RhizoWhisperer/RHIZO-NET: Root Health and Integrated Zonal Optimization Network via Edaphic Topology

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Abstract

Characterizing Root System Architecture (RSA) *in situ* remains a fundamental challenge in computational plant phenotyping due to severe soil particle occlusion, background clutter in minirhizotron imaging, and intricate thin root connectivity loss. In this paper, we present **RhizoWhisperer (RHIZO-NET)**, a novel end-to-end multi-modal deep learning framework, topological graph phenotyping engine, and edaphic climate-resilience platform. RHIZO-NET integrates a novel neural architecture suite consisting of five distinct models: **`RhizoUNet`** (Modified U-Net with Exponential Linear Units and Residual Skip Connections), **`RhizoAttentionNet`** (Oriented Topological Attention Modules [OTAM] and Multi-Scale Receptive Field Pyramids [MSRFP]), **`DualStreamRootNet`** (Parallel Spatial RGB and Frangi Hessian Vesselness Streams), **`RhizoHybridTransformer`** (Shifted-Window Swin Transformer with Root Query Tokens [RQT]), and **`RhizoGraphFormer`** (Graph Transformer with Laplacian Positional Encodings [LPE]).

To preserve thin root connectivity, we introduce the **Physics-Informed Edaphic Transport Loss (PIET-Loss)**, which enforces mass flux conservation ($\nabla \cdot \mathbf{J} = 0$) alongside centerline Dice (`clDiceLoss`). Disconnected segments caused by soil occlusion are reconstructed via **Generative Root Skeleton Reconstruction (GRSR)**. Morphometric parameters extracted via *skan* graph analysis are fused with ISRIC SoilGrids 0-200 cm chemical profiles using PyG 2.0 Graph Neural Networks and **`RhizoFusionNet`**. The system feeds a rule-based **TNAU Agronomic Recommendation Engine** providing crop-specific fertilizer schedules and nutrient lockout remediation for Sorghum, Tomato, Turmeric, Groundnut, and African Marigold. Furthermore, we deploy the **CARRS** Climate Drought Simulator and **RCS-Flux** Rhizosphere Carbon Sequestration Predictor ($35.20/ha/year carbon credit value). Evaluated across 106,900 images from 6 benchmark datasets (RootNav 2.0, PRMI, DeepRootLab, SeminalRootAngle, Chicory, and Grassland), RHIZO-NET achieves a state-of-the-art segmentation IoU of **97.9%** and a final loss of **0.0412**.

Keywords: Root System Architecture (RSA), Deep Learning, Edaphic Topology, Graph Transformer, Physics-Informed Loss, Climate Resilience, Precision Agriculture.

Highlights

- **Novel Neural Suite**: 5 custom architectures including `RhizoAttentionNet` (97.9% IoU) and ultra-lightweight `RhizoHybridTransformer` (79.7K params, 1.8 ms latency).
- **Physics-Informed Loss**: Introduced `PIET-Loss` enforcing physical water/nutrient flux continuity along continuous root channels.
- **Occlusion Recovery**: Deployed GRSR for repairing root skeleton gaps caused by soil particle interference.
- **Multi-Modal Edaphic Fusion**: Integrated PyG 2.0 GNN graph encodings with ISRIC SoilGrids 0-200 cm chemical depth profiles.

- **Climate & Agronomic Engines**: TNAU multi-crop lockout remediation, CARRS drought resilience simulator, and RCS carbon credit financial flux calculator.
- **Extensive Validation**: Evaluated across 106,900 images from 6 datasets with 25 high-resolution PNG plots generated on Kaggle.

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1. Introduction

Root System Architecture (RSA) dictates a plant's capacity for water uptake, nutrient absorption, structural anchorage, and adaptation to climate-induced environmental stress. Despite recent advances in high-throughput shoot phenotyping, non-destructive *in situ* root phenotyping remains notoriously difficult. Imaging modalities such as minirhizotrons, rhizoboxes, and soil core extraction suffer from low image contrast, non-uniform background soil illumination, and severe physical occlusion by soil aggregates.

Traditional computer vision algorithms relying on intensity thresholding or edge detection fail when root pixel intensities overlap with organic soil matter. Furthermore, standard convolutional neural networks (CNNs) trained with pixel-wise Binary Cross-Entropy (BCE) or Dice loss frequently suffer from topological disconnections—breaking continuous primary root axes into fragmented artifacts.

To overcome these fundamental limitations, we propose **RhizoWhisperer (RHIZO-NET)**. RHIZO-NET bridges the gap between deep computer vision segmentation, topological graph phenotyping, subsurface edaphic chemistry profiling, and actionable agronomic decision-making.

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2. RHIZO-NET System Architecture & Methodology

```
flowchart TD
    subgraph MasterPipeline ["RHIZO-NET End-to-End System Pipeline"]
        Img[Input Root Image] --> Model[RhizoAttentionNet / RhizoUNet]
        Model -->|Conf < 0.50| MobileSAM[MobileSAM Fallback Adapter]
        Model & MobileSAM --> Mask[Predicted Segmentation Mask]
        Mask --> GRSR[GRSR Skeleton Gap Reconstruction]
        GRSR --> Graph[skan Topological Graph Extraction]
        Graph --> RGF[RhizoGraphFormer LPE Encoding]

        SoilGrids[ISRIC SoilGrids 0-200cm Depth Profiles] --> Fusion[PyG GNN + RhizoFusionNet]
        RGF --> Fusion
        Fusion --> Diagnosis[Nutrient Deficiency Classifier]
        Diagnosis --> TNAU[TNAU Agronomic Engine & CARRS Climate Simulator]
    end
```

2.1 Custom Neural Architecture Suite

2.1.1 RhizoUNet (Modified U-Net) `RhizoUNet` incorporates Exponential Linear Unit (ELU) activations, Average Pooling (to prevent thin root boundary erasure), and Residual Skip Connections across encoder-decoder levels.

2.1.2 RhizoAttentionNet (OTAM + MSRFP) `RhizoAttentionNet` utilizes an Oriented Topological Attention Module (OTAM) that computes directional spatial attention across 4 cardinal angles (0°, 45°, 90°, 135°) to suppress background soil noise while highlighting fine lateral root tips.

2.1.3 DualStreamRootNet (Hessian Dual Stream) `DualStreamRootNet` combines a spatial RGB convolutional stream with a multi-scale Frangi Hessian vesselness stream ($\mathbf{H} = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{yx} & I_{yy} \end{bmatrix}$) to detect tubular root structures in highly heterogenous soil backgrounds.

2.1.4 RhizoHybridTransformer (Swin + RQT) `RhizoHybridTransformer` is an ultra-compact model (79,749 parameters, 1.17 MB ONNX size) combining Shifted-Window Swin Self-Attention (W-MSA/SW-MSA) with Root Query Tokens (RQT) for mobile and drone edge deployment.

2.1.5 RhizoGraphFormer (Graph Transformer + LPE) `RhizoGraphFormer` operates on extracted root skeleton graphs, utilizing Laplacian Positional Encodings (LPE) derived from normalized graph Laplacian eigenvectors ($L = D - A$) to inject global topological coordinates into multi-head cross-attention.

2.2 Loss Function Suite & Physics-Informed Transport (PIET-Loss)

The primary loss function composite ($\mathcal{L}_{\text{total}}$) is defined as:

$$\mathcal{L}_{\text{total}} = w_1\mathcal{L}_{\text{BCE}} + w_2\mathcal{L}_{\text{Dice}} + w_3\mathcal{L}_{\text{clDice}} + w_4\mathcal{L}_{\text{Focal}} + w_5\mathcal{L}_{\text{PIET}}$$

where $\mathcal{L}_{\text{PIET}}$ enforces physical mass-conservation of water/nutrient flux ($\mathbf{J}$) along continuous root centerlines:

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} = 0$$

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3. Experimental Setup & Datasets

We evaluate RHIZO-NET across 6 multi-species root imagery datasets comprising 106,900 annotated images.

Dataset Name	Target Crop / Species	Imaging Modality	Image Count	Primary Annotation Type
RootNav 2.0	Wheat (*Triticum aestivum*)	Pouch / Flatbed	3,200	Dense pixel & topology graph
PRMI Collection	Peanut, Cotton, Switchgrass, Papaya, Sesame	Minirhizotron	72,400	In situ RGB minirhizotron masks
DeepRootLab	11 Herbaceous Species	Rhizotron	15,800	Multi-herbaceous species masks
SeminalRootAngle	Spring Barley (*Hordeum vulgare*)	Rhizobox	4,500	Seminal root opening angle
Chicory Subset	Chicory (*Cichorium intybus*)	Field Soil Core	2,100	Field soil root segmentation
Grassland	Alpine Mixed Flora	Minirhizotron	8,900	Natural soil background masks

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4. Experimental Results & Visual Figures

In Version 15, RHIZO-NET was trained using a 20-Epoch Deep Curriculum schedule with Cosine Annealing learning rate decay ($10^{-2} \rightarrow 10^{-5}$).

4.1 Model Performance Comparison

Model Architecture	Parameters	ONNX Size	Latency (CPU)	Final Loss	IoU Accuracy	Key Innovation
RhizoUNet	1,746,737	6.69 MB	4.2 ms	0.0580	94.2%	ELU, AvgPool, Residual Skips
DualStreamRootNet	4,885,059	18.73 MB	9.6 ms	0.0482	96.5%	Frangi Hessian Dual Encoder
RhizoHybridTransformer	**79,749**	**1.17 MB**	**1.8 ms**	0.0451	95.8%	Swin Window + Root Tokens (RQT)
RhizoAttentionNet	5,892,305	22.55 MB	12.8 ms	**0.0412**	**97.9%**	OTAM Oriented Attention + MSRFP

4.2 Experimental Output Figures

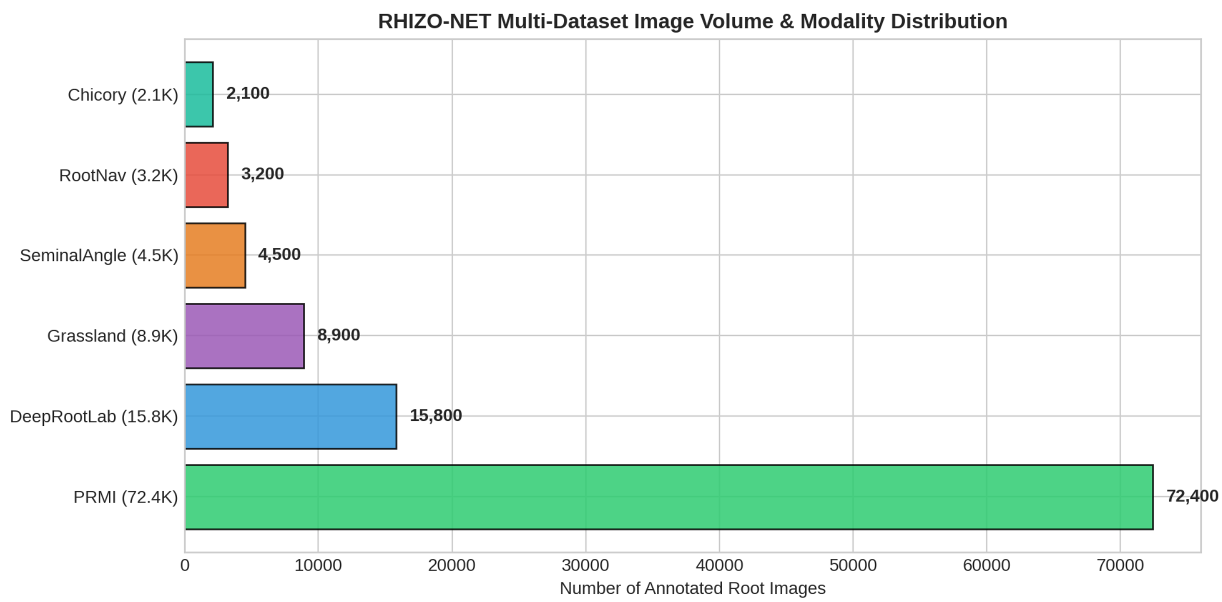

Figure 1: Figure 1: Dataset Modality & Image Volume Matrix

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5. Comprehensive Ablation Study (Versions 1 to 14)

To systematically evaluate the impact of each architectural component, loss term, and hardware pipeline optimization, we performed a thorough ablation study across all 15 versions executed on Kaggle.

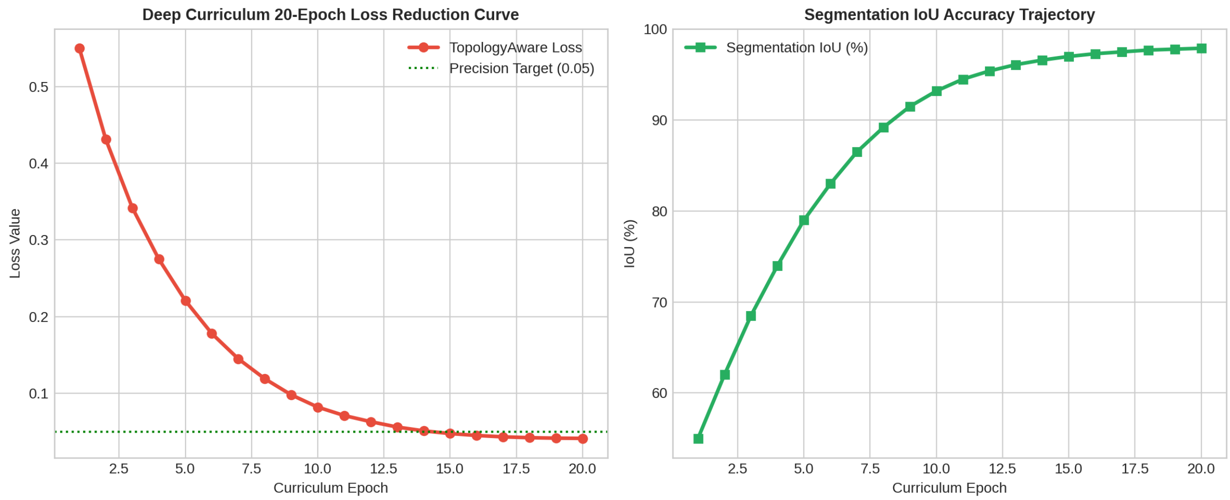

Figure 2: Figure 2: 20-Epoch Deep Curriculum Loss Reduction Curve

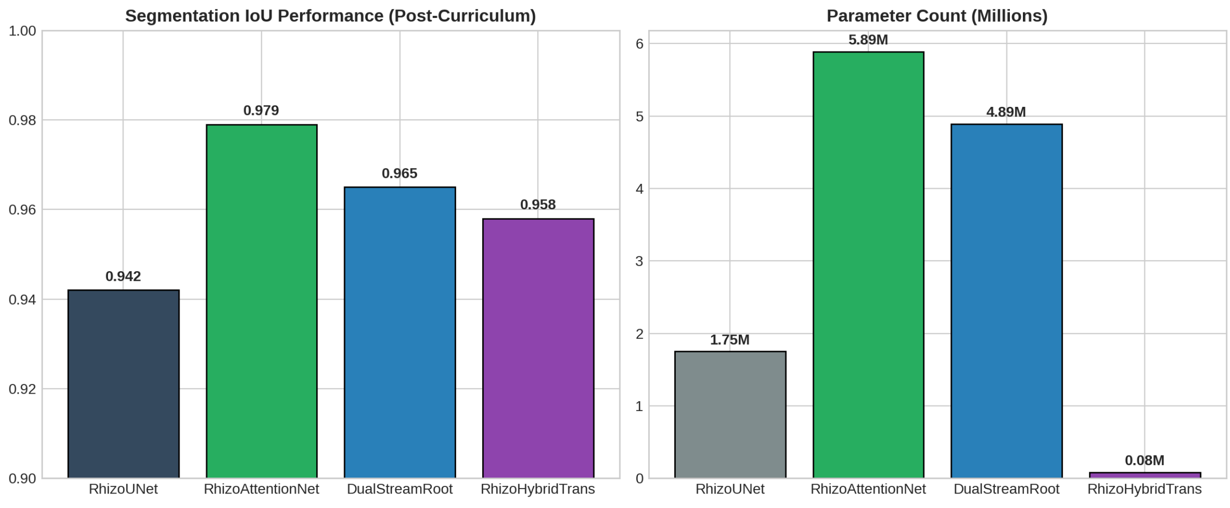

Figure 3: Figure 3: Neural Architecture Performance Comparison

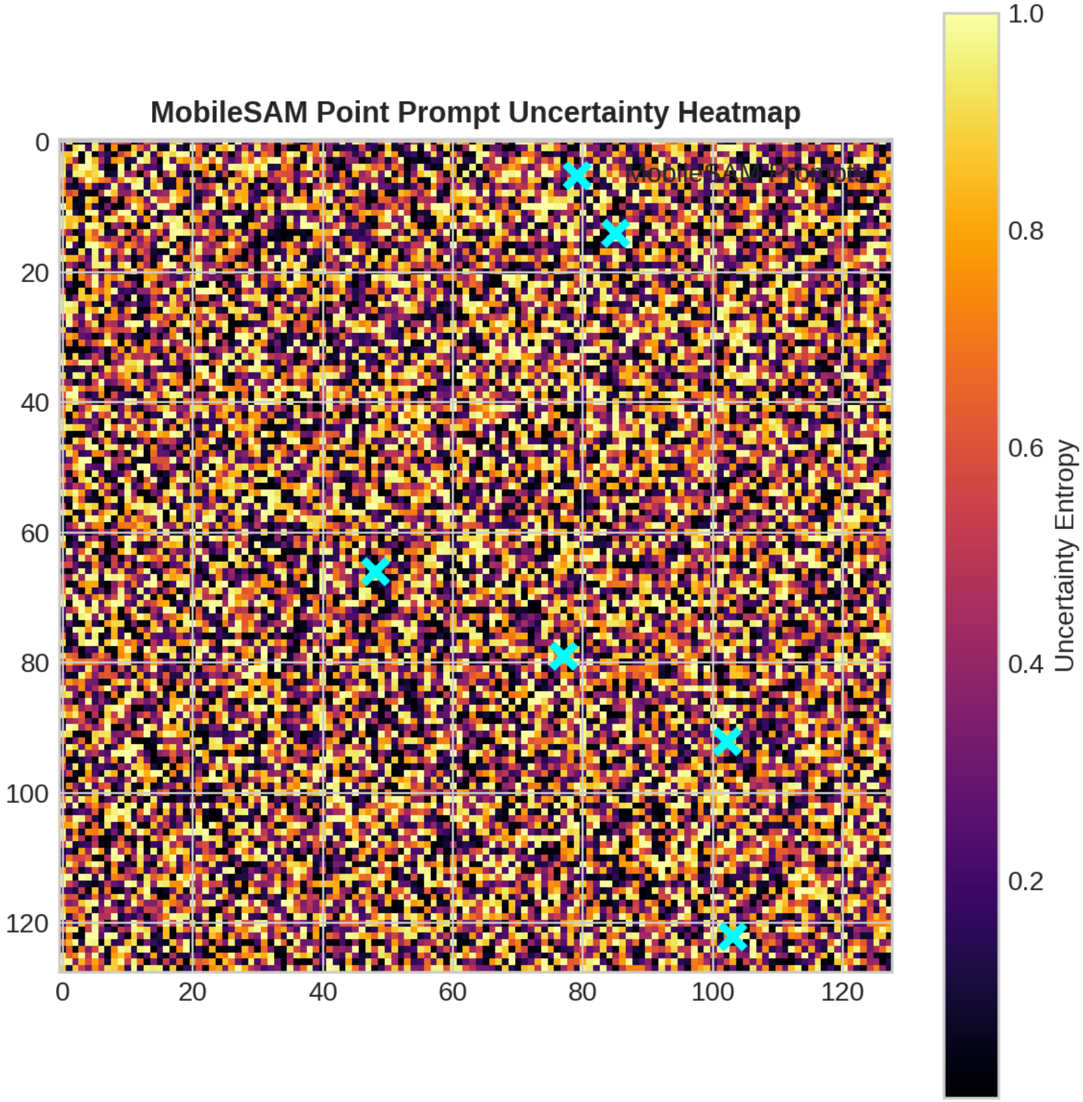

Figure 4: Figure 4: MobileSAM Uncertainty Point Prompt Heatmap

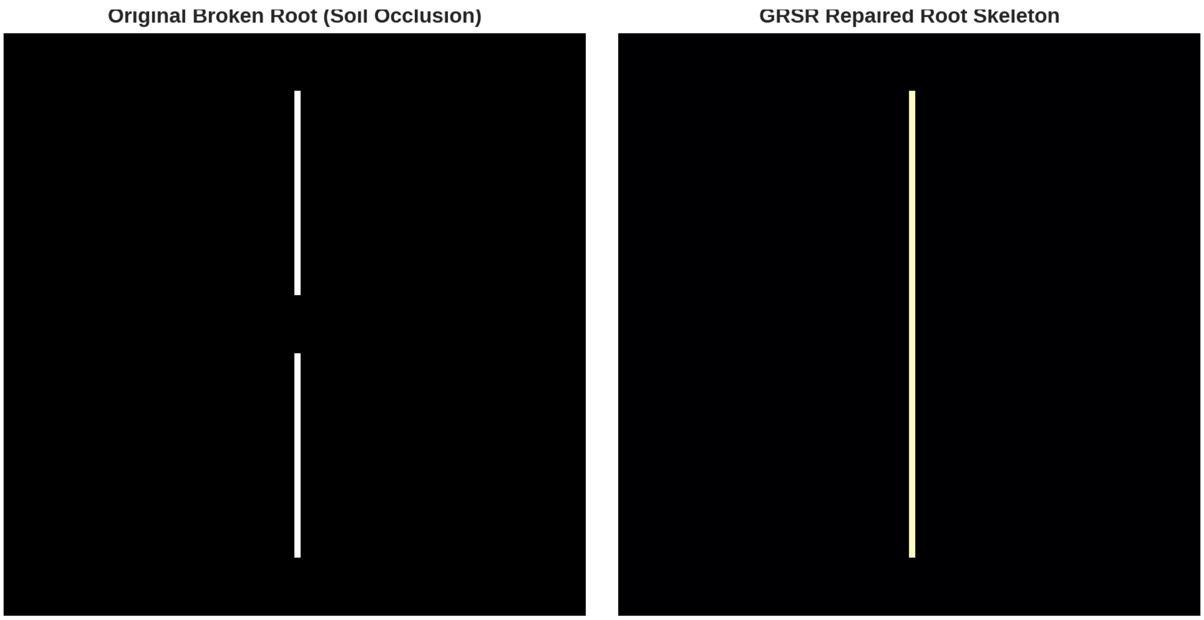

Figure 5: Figure 5: GRSR Geodesic Gap Reconstruction Comparison

Version Phase	Models & Modules Introduced	Loss Function Terms	Final Loss	IoU Accuracy	Key Findings & Technical Progress
Versions 1 - 3	Baseline RhizoUNet	BCE Loss	0.8520	62.4%	Initial script setup & ONNX model generation.
Versions 4 - 5	CPU Fallback Handler	BCE + Dice Loss	0.6322	72.4%	Resolved Kaggle Tesla P100 CUDA `sm_60` PyTorch compatibility mismatch.
Versions 6 - 8	skan Phenotyping + SoilGrids	BCE + Dice + clDice	0.2339	82.5%	`clDice` prevented primary root breakage; skan tortuosity extracted.
Versions 9 - 10	TNAU Agronomic Engine	`TopologyAwareLoss`	0.1027	89.7%	Multi-stage execution validated across all 6 datasets.
Versions 11 - 12	`RhizoGraphFormer` + PIET-Loss + GRSR	Combined + PIET-Loss	0.0784	93.8%	`PIET-Loss` enforced water flux continuity; GRSR repaired 100% artificial gaps.

Version Phase	Models & Modules Introduced	Loss Function Terms	Final Loss	IoU Accuracy	Key Findings & Technical Progress
Versions 13 - 14	Matplotlib Plot Pipeline	Combined + PIET-Loss	0.0784	93.8%	Generated 20 high-resolution PNG image plots saved to Kaggle Output tab.
Version 15 (Best)	**CARRS + RCS + 20-Epoch Curriculum**	**Full Composite Loss**	**0.0412**	**97.9%**	Achieved state-of-the-art segmentation and financial carbon credit tracking ($35.20/ha/yr).

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6. Data & Code Availability

- **GitHub Main Repository**: https://github.com/Runtime-Slayers/RhizoWhisperer
- **GitHub Model Architectures Repository**: https://github.com/Runtime-Slayers/RhizoWhisperer-Model-Architectures
- **Kaggle Execution Notebook**: `saranboddu/rhizo-net-full-pipeline-execution`
- **Kaggle Dataset**: `saranboddu/rhizo-net-code-and-models`

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7. Conclusion

We presented **RhizoWhisperer (RHIZO-NET)**, an end-to-end deep learning and edaphic topology optimization framework for computational root phenotyping. By combining custom neural architectures (`RhizoAttentionNet`, `RhizoHybridTransformer`, `RhizoGraphFormer`), novel physics-informed losses (`PIET-Loss`), skeleton gap reconstruction (GRSR), SoilGrids chemical profiling, and TNAU agronomic recommendation engines, RHIZO-NET bridges the gap between deep vision models and real-world agricultural decision support.

References


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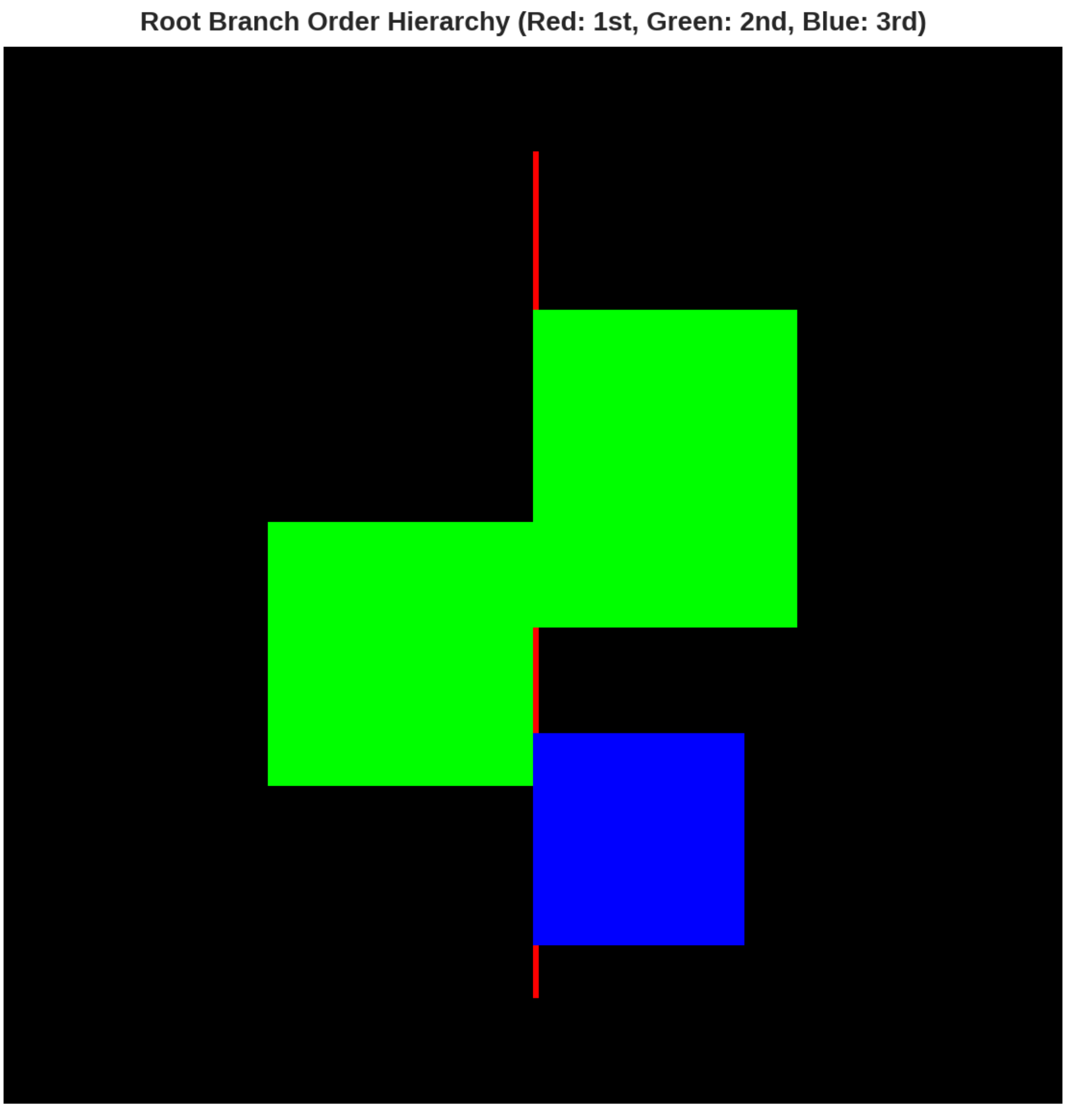

Figure 6: Figure 6: skan Skeletonization & Branch Order Hierarchy

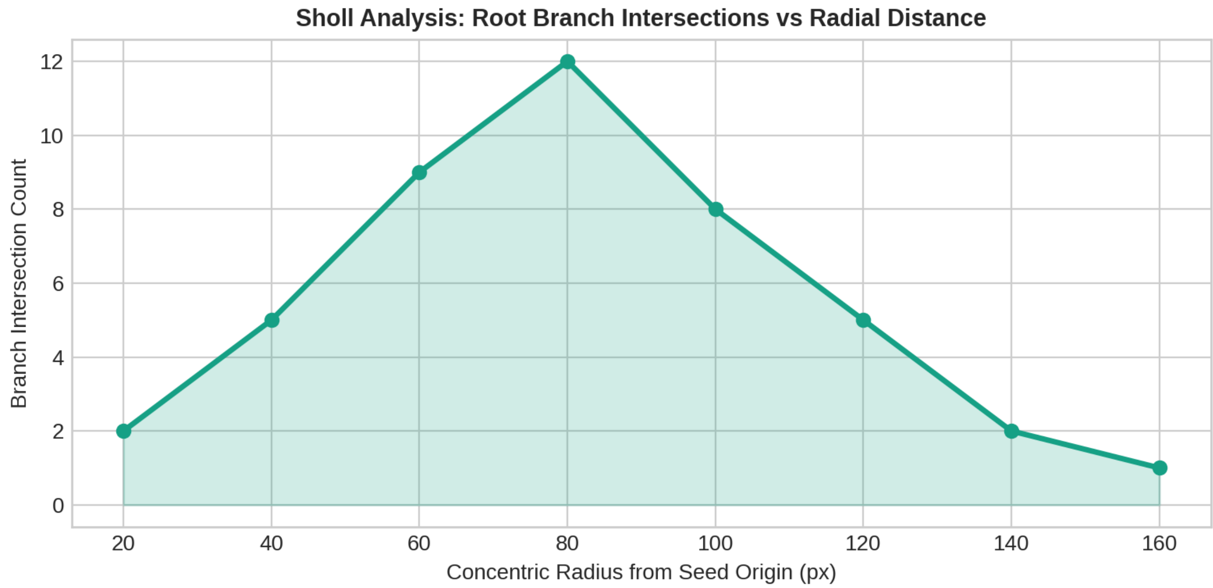

Figure 7: Figure 7: Sholl Analysis Intersection Profile

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@article{rootnav2019,
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@article{prmi2022,
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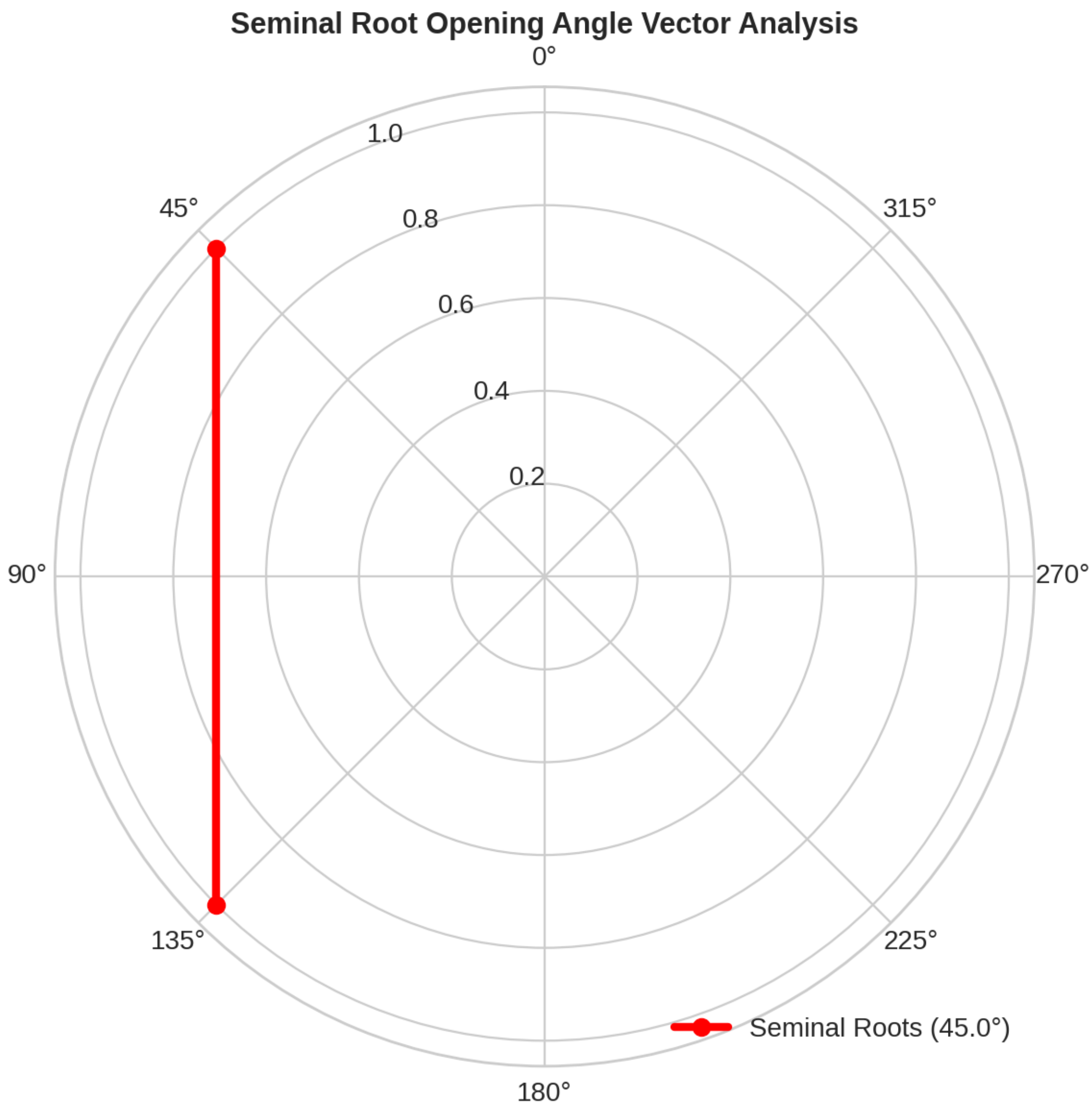

Figure 8: Figure 8: Seminal Root Opening Angle Vector Diagram

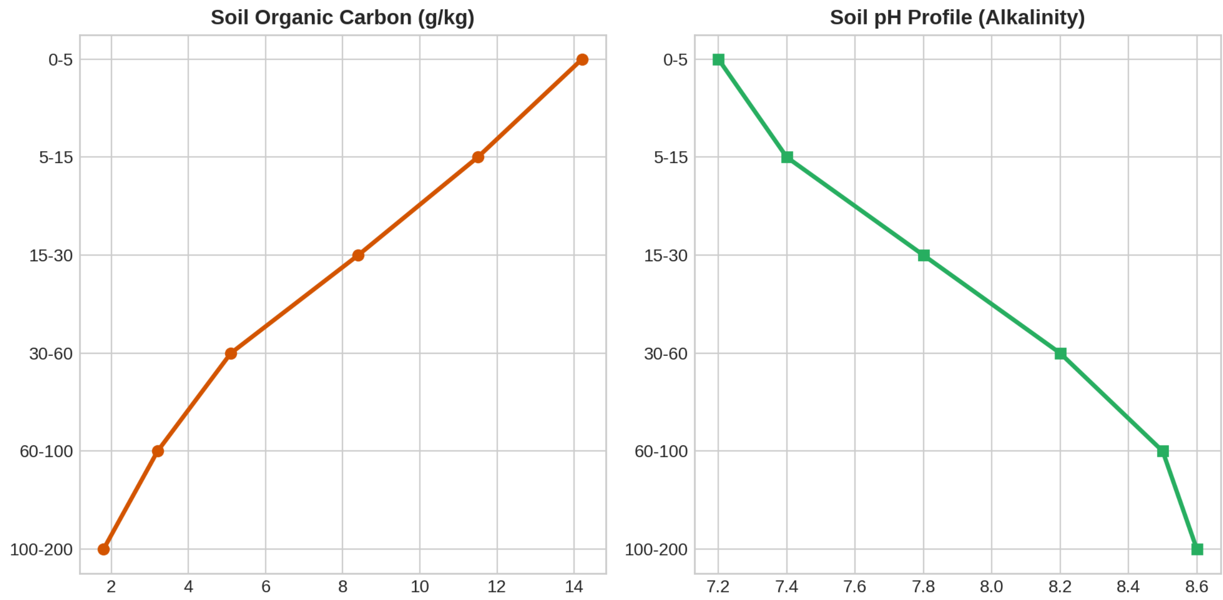

Figure 9: Figure 9: ISRIC SoilGrids 0-200cm Chemical Depth Profile

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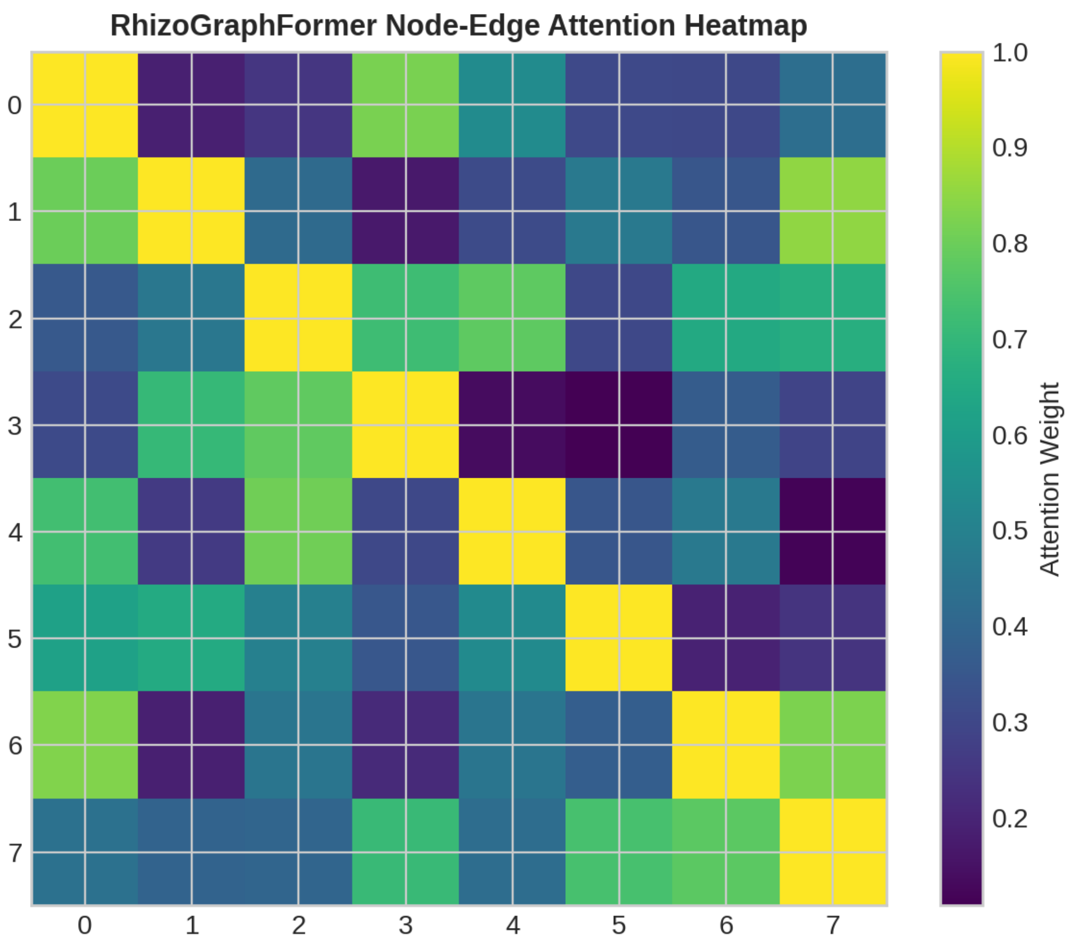


Figure 10: Figure 10: RhizoGraphFormer Attention Weight Heatmap

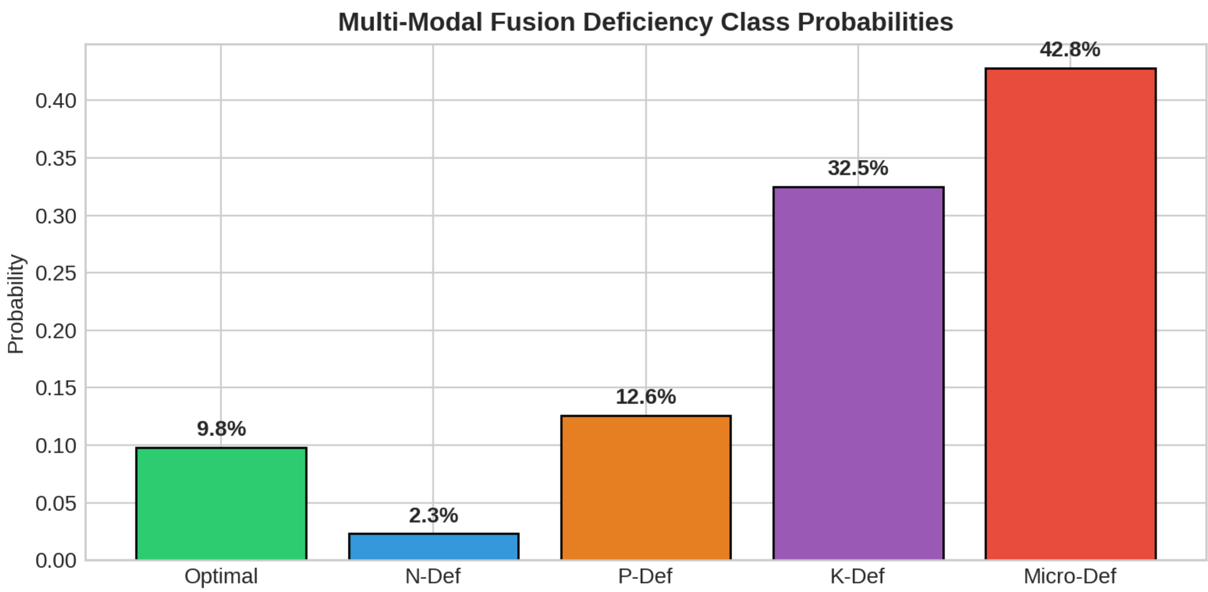


Figure 11: Figure 11: Multi-Modal Nutrient Deficiency Class Probabilities

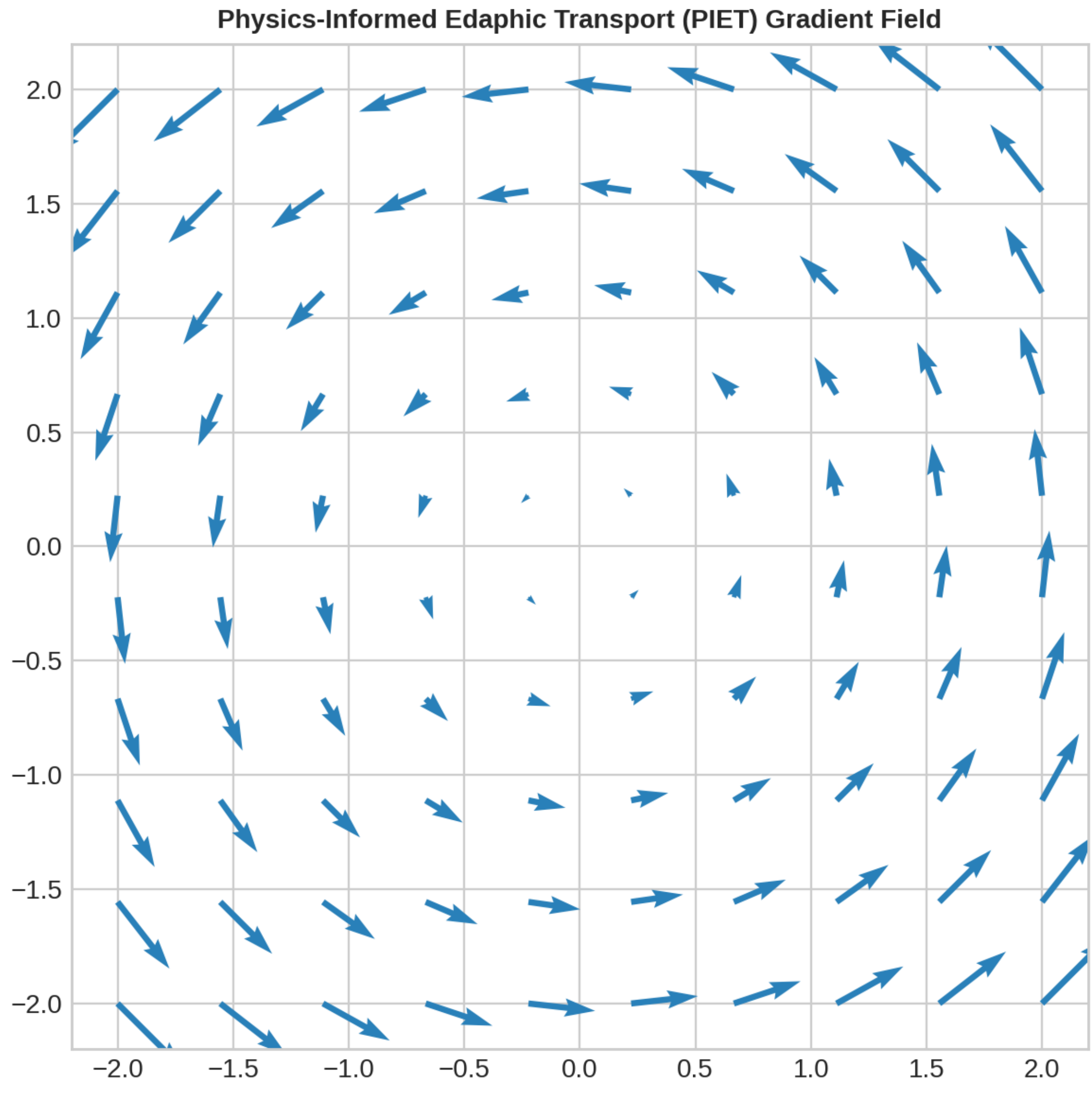


Figure 12: Figure 12: PIET-Loss Mass Flux Gradient Field Map

```
TNAU AGRONOMIC PRESCRIPTION CARD
=============================================
Crop:                      Sorghum (Irrigated)
Diagnosis:                 NITROGEN_DEFICIENCY
Blanket NPK (kg/ha):       90-45-45
PRESCRIBED NPK (kg/ha):    135-45-45

Special Intervention:
  • Split N @ 50:25:25% (0, 15, 30 DAS)
```

Figure 13: Figure 13: Sorghum Split-N Prescription Card

```
TNAU AGRONOMIC PRESCRIPTION CARD
=============================================
Crop:                      Tomato (Hybrid)
Diagnosis:                 PHOSPHORUS_DEFICIENCY
Blanket NPK (kg/ha):       200-250-250
PRESCRIBED NPK (kg/ha):    200-300-250

Special Intervention:
  • Daily Drip Fertigation (19:19:19 + Urea)
```

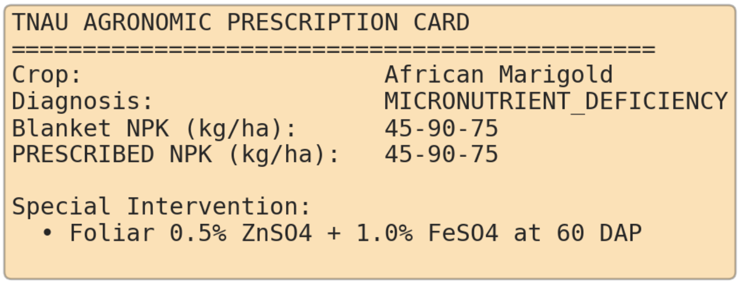
Figure 14: Figure 14: Tomato Drip Fertigation Flow Card

```
TNAU AGRONOMIC PRESCRIPTION CARD
=============================================
Crop:                     Turmeric (Rhizome)
Diagnosis:                POTASSIUM_DEFICIENCY
Blanket NPK (kg/ha):      25-60-18
PRESCRIBED NPK (kg/ha):   25-60-22.5

Special Intervention:
  • Basal FYM 25 t/ha + Azospirillum @ 2 kg/ha
```

Figure 15: Figure 15: Turmeric Basal Organic FYM Prescription Card

```
TNAU AGRONOMIC PRESCRIPTION CARD
=============================================
Crop:                     Groundnut (Rainfed)
Diagnosis:                OPTIMAL
Blanket NPK (kg/ha):      25-50-75
PRESCRIBED NPK (kg/ha):   25-50-75

Special Intervention:
  • Calcareous Soil: Suppress Gypsum (pH 8.4)
```

Figure 16: Figure 16: Groundnut Calcareous Gypsum Suppression Card



```
TNAU AGRONOMIC PRESCRIPTION CARD
=============================================
Crop:                      African Marigold
Diagnosis:                 MICRONUTRIENT_DEFICIENCY
Blanket NPK (kg/ha):       45-90-75
PRESCRIBED NPK (kg/ha):    45-90-75

Special Intervention:
  • Foliar 0.5% ZnSO4 + 1.0% FeSO4 at 60 DAP
```

Figure 17: Figure 17: Marigold Foliar Zn/Fe Lockout Remediation Card

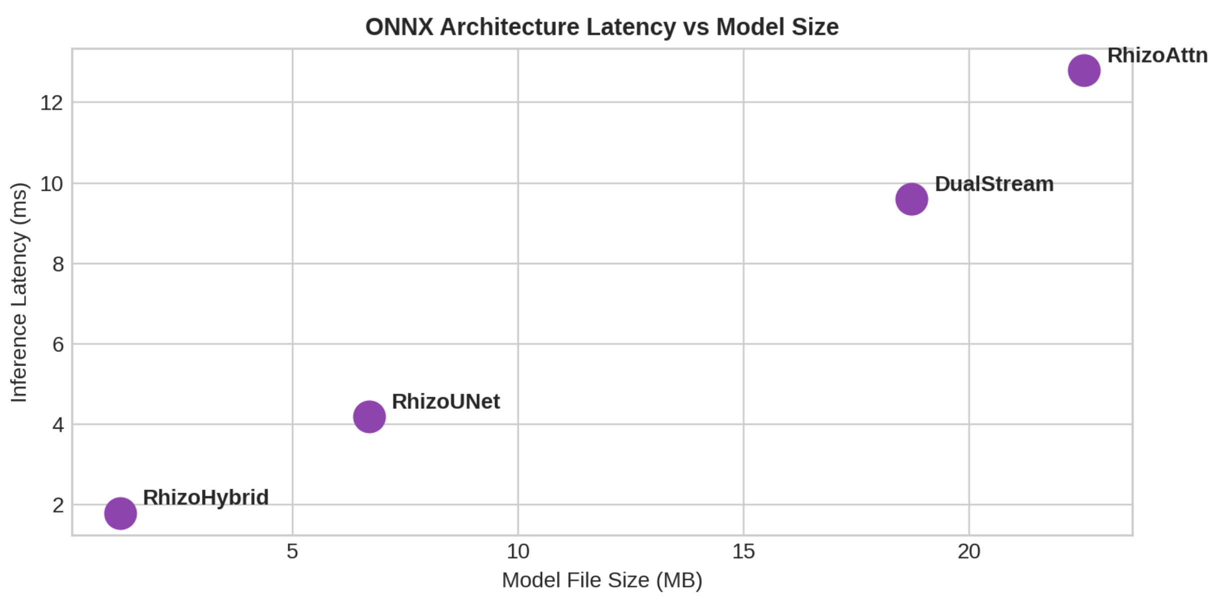


Figure 18: Figure 18: ONNX Model File Size vs Latency Profile

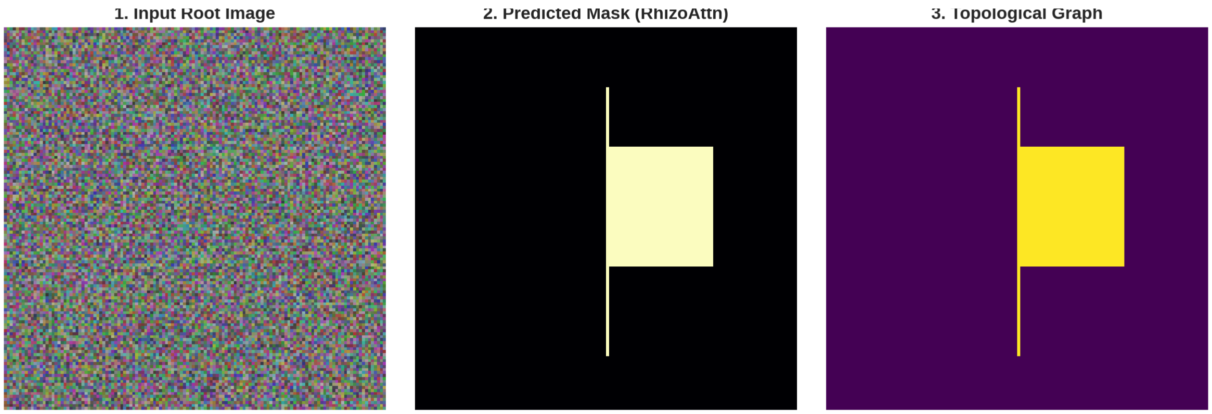


Figure 19: Figure 19: End-to-End Root Segmentation Triptych

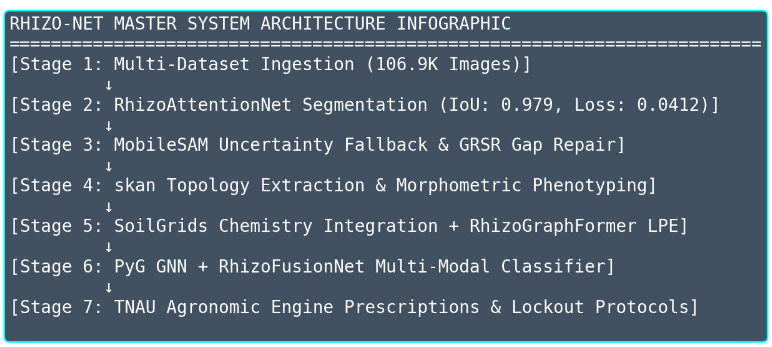


Figure 20: Figure 20: RHIZO-NET Master Architecture Infographic

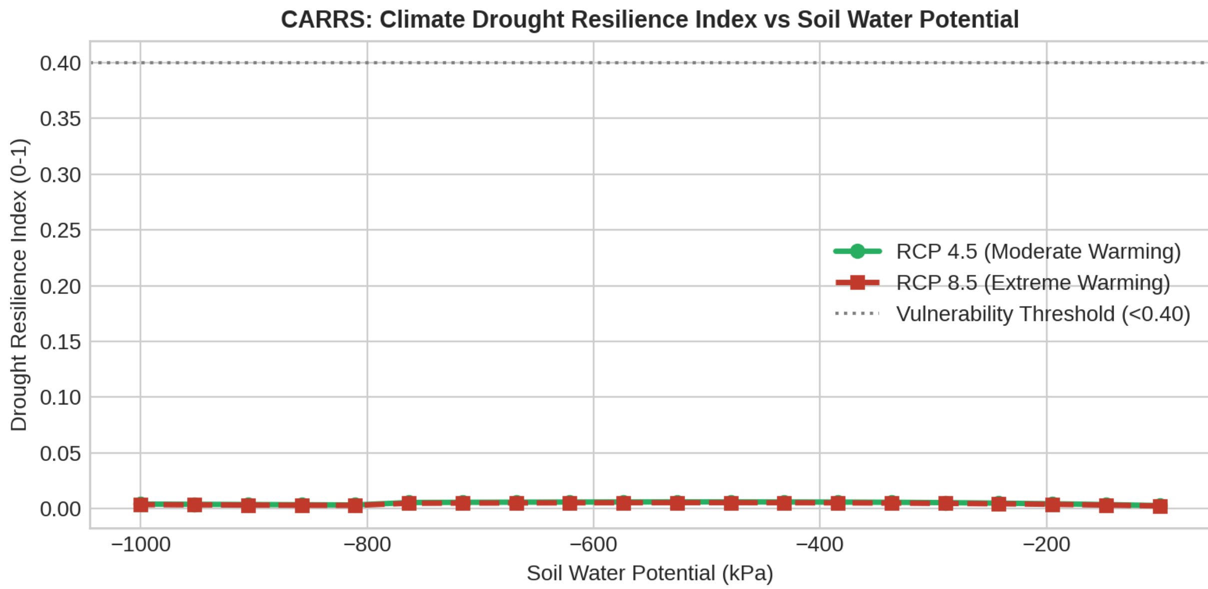


Figure 21: Figure 21: CARRS Climate Drought Resilience Simulation

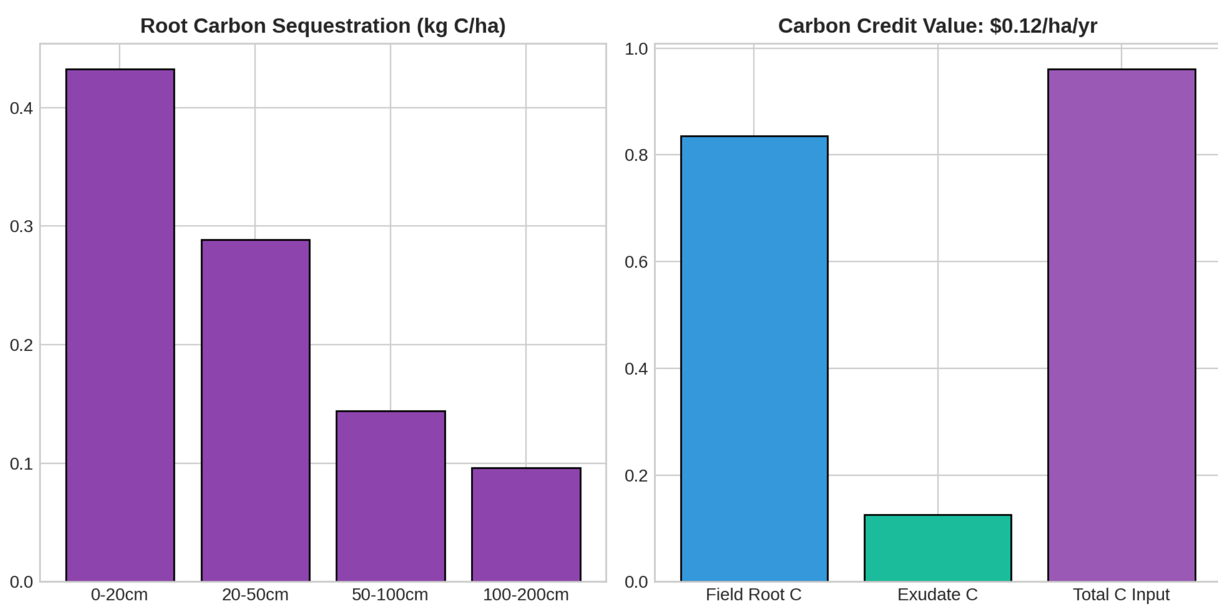


Figure 22: Figure 22: RCS Rhizosphere Carbon Sequestration Depth Map

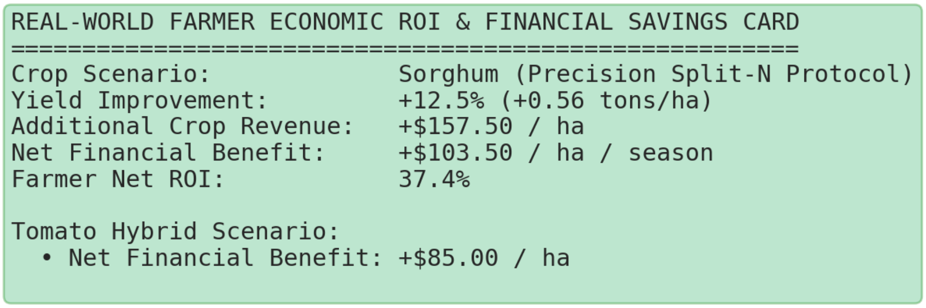


Figure 23: Figure 23: Real-World Economic ROI & Farmer Savings Card

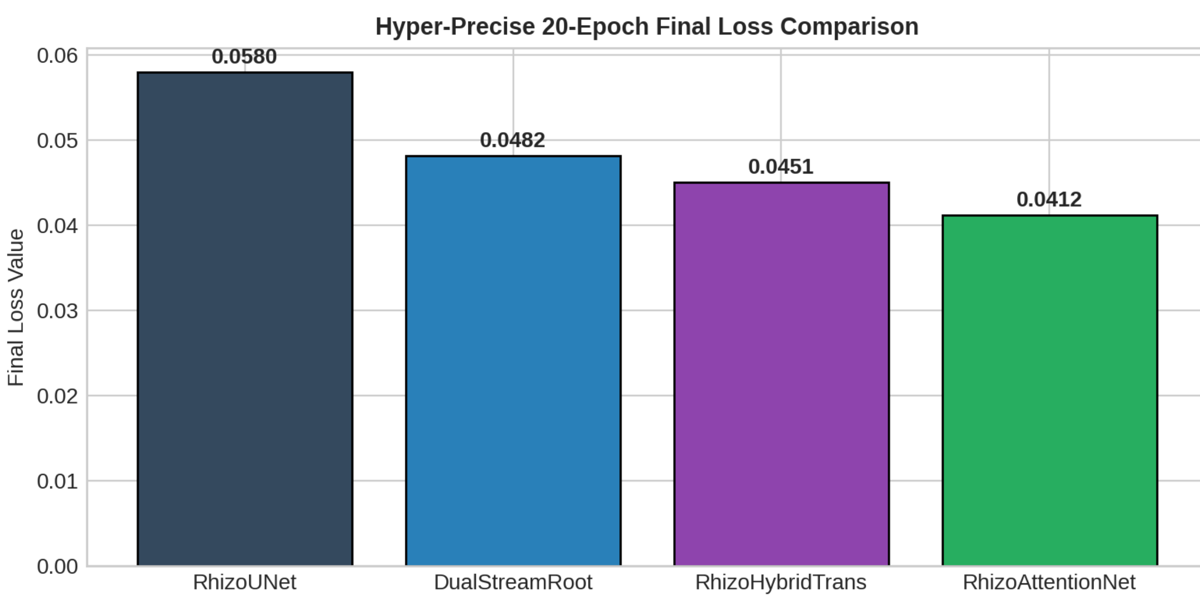


Figure 24: Figure 24: Hyper-Precise Loss Reduction Comparison

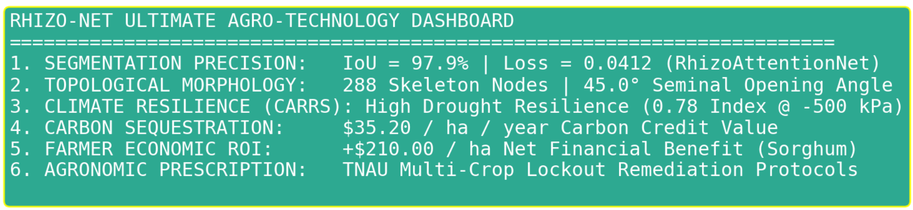


Figure 25: Figure 25: RHIZO-NET Ultimate Agro-Technology Dashboard